

PROJECT REPORT

Is Europe's Energy Transition Actually Working?

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1 INTRODUCTION AND QUESTION

Over the past three decades, European countries have committed to shifting away from fossil fuels toward renewable energy. Political agreements, carbon targets, and green investment pledges have multiplied – but is the transition actually happening at the pace needed, and is it happening evenly across the continent?

This project explores energy mix data from 38 European countries between 1990 and 2024, combining Eurostat energy balance data with CO₂ emissions records from Our World in Data. The goal is to reveal who is leading the transition, who is falling behind, and what the aggregate numbers hide about individual country trajectories.

The dataset covers not just EU member states but also the UK, Norway, Iceland, the Western Balkans, Ukraine, Moldova and Georgia – giving a fuller picture of the broader European region and allowing meaningful comparisons between countries inside and outside the EU.

2 DATA

This study utilizes a longitudinal synthesis of macroeconomic and environmental indicators to evaluate the decoupling of economic growth from CO₂ emissions across 38 European nations from 1990 to 2024. The research framework integrates the Our World in Data (OWID) CO₂ dataset for production-based emissions and GDP (analyzed at purchasing power parity in constant international dollars) with the Eurostat Simplified Energy Balances (NRG_BAL_S), which provides granular energy consumption data in Thousand Tonnes of Oil Equivalent (KTOE). A relational mapping was established to resolve geopolitical coding discrepancies between Eurostat's 2-letter identifiers and ISO-3 alpha codes, enabling the engineering of critical analytical metrics such as the renewable penetration rate (`renewable_share_pct`), fossil fuel reliance (`fossil_share_pct`), and the carbon intensity of energy (`co2_per_unit_energy`). While the dataset provides a robust basis for efficiency evaluation, its reliance on production-based metrics necessitates a cautious interpretation regarding "carbon leakage" in service-heavy economies, and fragmented records in early 1990s Eastern European data were addressed through targeted exclusions to maintain the statistical significance of the observed decoupling trajectories.

3 AUDIENCE, TASKS AND DESIGN RATIONALE

The system is specifically designed for a general audience interested in energy and climate policy, ranging from students and researchers to environmental activists and journalists who require clear, visual evidence to evaluate Europe's energy transition. To support this, the platform addresses five core analytical tasks: T1, comparing renewable energy shares across European nations; T2, tracking shifts in a country's energy mix from 1990 to 2024; T3, identifying which countries are effectively reducing CO₂ emissions; T4, exploring the correlation between renewable adoption and per capita emissions; and T5, analyzing the gap between "green pledges" and actual fossil fuel dependency. The design rationale focuses on making these complex datasets intuitive and comparable; for instance, comparative charts are used for T1, T3, and T5 to immediately highlight leaders and laggards in the "green race," while time-series visualizations - such as line or area charts – are employed for T2 to demonstrate long-term trends and the gradual displacement of fossil fuels. By utilizing scatter plots for T4 to reveal the direct relationship between renewable investment and environmental impact, the system provides a cohesive visual narrative that helps users answer the project's central question: whether the energy transition is actually working.

4 SYSTEM DESCRIPTION

The system is architected as a cohesive WordPress-hosted analytical platform that integrates multiple visualization layers through a synchronized interactive framework. At the core of the interface, a Choropleth Map of Europe serves as the primary navigational filter, where the selection of a specific geopolitical entity triggers coordinated cross-filtering across all secondary views, including a Decoupling Scatter Plot. This seamless integration allows for the immediate transition from geospatial observation to causal analysis, supported by a global temporal slider spanning 1990 to 2024 that enables users to visualize the longitudinal evolution of energy policies and economic shifts. A core feature is the custom-engineered "Country Energy & CO₂ Explorer" developed in java, which performs dynamic real-time calculations to adapt the interface to any selected nation and year. Beyond visualizing the energy mix (renewable vs. fossil shares), this component provides an automated, actionable summary that benchmarks a country's CO₂ emissions against European averages. By bridging the gap between raw longitudinal data and comparative analysis, the system enables users to evaluate national performance within the broader continental context from 1990 to the present day.

5 REJECTED ALTERNATIVES

The final design of the system was the result of an iterative process where several visualization strategies were evaluated and subsequently dismissed to prioritize analytical clarity. First, the inclusion of a Stacked Area Chart for the aggregate energy mix of all 38 countries was rejected; while effective for showing total volume, it created a significant "spaghetti effect" that obscured individual country trajectories and made regional comparisons cognitively taxing. To resolve this, the design pivoted toward an interactive Choropleth Map linked to specific country views, which preserved geographic context without overwhelming the user. Second, the use of Absolute CO₂ Tonnage as the primary metric was reconsidered and replaced by CO₂ per capita and Carbon Intensity. Using raw totals unfairly penalized large economies and failed to account for population growth or industrial scale, whereas normalized metrics allowed for a more scientifically rigorous evaluation of efficiency gains and genuine decoupling progress. These adjustments ensured that the final system favored relative performance and structural trends over misleading absolute values.

6 FINDINGS AND LIMITATIONS

The analytical synthesis of the European energy transition reveals that while significant progress has been achieved, it remains geographically uneven and structurally complex. Key insights indicate a clear leadership tier, with Iceland (83.6%), Norway (56.7%), and Sweden (48.4%) achieving the highest renewable shares driven by robust policy frameworks and natural resource advantages; conversely, "persistent laggards" such as Malta and Cyprus remain over 85% dependent on fossil fuels. A critical finding is the strong negative correlation ($r = -0.52$) between renewable adoption and CO₂ per capita, empirically validating that higher renewable integration effectively drives down emissions. However, ethical reflection through the White-Hat/Black-Hat lens reveals that production-side successes often mask "outsourced emissions," where countries like Luxembourg and Belgium appear green by effectively exporting their manufacturing footprint to other regions. This highlights a fundamental limitation of the current model: the reliance on production-based metrics may offer a deceptive narrative of decoupling.

Future improvements should therefore focus on integrating consumption-based CO₂ data to provide a more transparent global accountability model, as current trajectories suggest that despite individual successes like Germany's 49% reduction, the overall pace of the EU's 2030

climate targets remains insufficient. By expanding the system to include real-time sectoral data and predictive modeling, the visual framework could more accurately address the gap between national green pledges and actual fossil fuel dependency.